1) $4^3 = 4 \cdot 4 \cdot 4 = 64$

2) $(-2)^4 = (-2)\cdot(-2)\cdot(-2)\cdot(-2) = 16$

3) $(-5)^3 = (-5)\cdot(-5)\cdot(-5) = -125$

4) $25^0 = 1$

5) $(-9)^0 = 1$

6) $\left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

7) $-2014^0 = -1$

8) $(-2)^{-5} = \left(-\frac{1}{2}\right)^5 = -\frac{1}{32}$

9) $-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -16$

10) $-4^5 + 3\cdot 4^5 - 7\cdot 3^4 - 3\cdot 3^4 - 4^5 = 4^5(-1+3-1) + 3^4\cdot(-7-3)$
$= 4^5 \cdot 10 \cdot 3^4$

11) $\frac{5^{80}+5^{85}+5^{80}}{5^{85}+5^{80}+5^{75}} = x$ $\quad \frac{x}{125} = ?$ $\quad \frac{5^5}{5^3} = 5^2 = 25$

$\frac{5^{80}(1+5+10)}{5^{75}(5^5+5+1)} = 5^{80-75} = 5^5$

12) $\frac{14^n + 6^n + 2^n}{7^n + 3^n + 1} = ?$ $\quad \frac{2^n(7^n+3^n+1)}{7^n+3^n+1} = 2^n$

13) $(0,027)^{\frac{5}{3}} \cdot 10^5 = ?$ $\quad (27\cdot 10^{-3})^{\frac{5}{3}} \cdot 10^5 = (3^3)^{\frac{5}{3}} \cdot (10^{-3})^{\frac{5}{3}} \cdot 10^5$

$= 3^5 \cdot 10^5 \cdot 10^5 = 3^5 \cdot 10^0 = 3^5 \cdot 1 = 3^5$

14) $\frac{-9^{-2}\cdot 25^{-1}}{125^{-1}\cdot 27^{-1}} = ?$ $\quad \frac{-3^{-4}\cdot 5^{-2}}{5^{-3}\cdot 3^{-3}} = -\frac{5^1}{3^1} = -\frac{5}{3}$

15) $\frac{x^{a+1}}{x^{a-2}} \cdot \left[\frac{x^{-a+2}\cdot x^{2a+3}}{x}\right]^{-1} = ?$ $\quad x^{a+1-a+2}\left(x^{-a+2+2a+3-1}\right)^{-1}$

$= x^3 (x^{a+4})^{-1} = x^{3-a-4} = x^{-a-1}$

16) $\left(\frac{a^x}{a^y}\right)^{x-y} \cdot \left[\frac{a^b}{a^x}\right]^{x-y} = ?$ $\quad = (a^{x-y})^{x-y} \cdot (a^{y-x})^{x-y}$

$= (a^{x-y+y-x})^{x-y} = (a^0)^{x-y} = 1^{x-y} = 1$

17) $3^x = 2$ $\quad \frac{27^{x+1}}{9^{x-1}} - 81^x = ?$ $\quad (3^x)^3 \cdot 3^3 - (3^x)^4$

$(3^3)^{x+1} - (3^4)^x$ $\quad 2^3 \cdot 3^3 - 2^4$

$= 200$

18) $x-2=y \quad |x-y|+|y-x| = ?$

$|2| + |-2| = 4$

19) $-2<x<4 \quad |x+2|+|x-4| = ?$

$(x+2)-(x-4) = (x+2)-x+4 = 6$

20) $x<0<y \quad \frac{|2x-y|-|2y-x|}{x+y} = ?$

$\frac{-(2x-y)-(2y-x)}{x+y} = \frac{-2x+y-2y+x}{x+y} = \frac{-(x+y)}{x+y} = -1$

21) $\sqrt{0,64}+\sqrt{0,04}-\sqrt[3]{0,027} = \sqrt{\frac{64}{100}}+\sqrt{\frac{4}{100}}-\sqrt[3]{\frac{27}{1000}}$

$\sqrt{\left(\frac{8}{10}\right)^2}+\sqrt{\left(\frac{2}{10}\right)^2}-\sqrt[3]{\left(\frac{3}{10}\right)^3} = \frac{8}{10}+\frac{2}{10}-\frac{3}{10} = \frac{7}{10}$

22) $\frac{\sqrt{75}-\sqrt{27}+\sqrt{12}}{\sqrt{108}+\sqrt{48}} = \frac{5\sqrt{3}-3\sqrt{3}+2\sqrt{3}}{6\sqrt{3}+4\sqrt{3}} = \frac{4\sqrt{3}}{10\sqrt{3}} = \frac{2}{5}$

23) $\sqrt{2}(\sqrt{18}+\sqrt{50}-\sqrt{32}+\sqrt{8}) = \sqrt{36}+\sqrt{100}-\sqrt{64}+\sqrt{16}$

$= 6+10-8+4$

$= 12$

24) $\frac{\sqrt[5]{16}}{\sqrt[5]{\frac{1}{2}}} = \sqrt[5]{32} = \sqrt[5]{2^5} = 2$

25) $\sqrt{25-25x^2}-\sqrt{64-64x^2} = 5\sqrt{1-x^2}-8\sqrt{1-x^2}$

$= -3\sqrt{1-x^2}$

26) $\sqrt[5]{\sqrt{3\sqrt{2}}} = \sqrt[120]{2} = \sqrt[120]{2}$

27) $\sqrt[3]{2^3\sqrt{2^4}} = \sqrt[3]{2^4} = 2^{\frac{4}{3}}$

28) $a>0 \quad (\sqrt[5]{a^2})^4 = ?$

$\sqrt[5]{a^8} = \sqrt[5]{a^5 \cdot a^3} = a\sqrt[5]{a^3}$

29) $\sqrt{(-4)^2}+\sqrt[3]{(-2)^3}+\sqrt[4]{(-2)^4}+\sqrt[4]{3^4} = |-4|-(-2)+|-2|+3 = 11$

30) $a<0$

32) $\frac{2}{\sqrt{6}}$

34) A

$\frac{\sqrt{5}}{\sqrt{5}}$

30) $a<0$ $\sqrt{a^2}+\sqrt{(-a)^2}-\sqrt{4a^2}=?$

$|a|+|-a|-|2a|$

$-a+(-a)-(-2a$

$=-2a+2a=0$

31) $a>0, b<0$ $\sqrt{(b-a)^2}-\sqrt{(2a-b)^2}$

$-(b-a)-(2a-b)$

$=-b+a-2a+b=-a$

32) $\frac{2}{\sqrt{6}}=?$ $\frac{2}{\sqrt{6}}=\frac{2\sqrt{6}}{6}=\frac{\sqrt{6}}{3}$

$(\sqrt{6})$

33) $\frac{2\sqrt{3}}{\sqrt{5}-\sqrt{3}}=\frac{2\sqrt{3}(\sqrt{5}+\sqrt{3})}{\sqrt{5}^2-\sqrt{3}^2}$

$\frac{2(\sqrt{15}+3)}{2}=\sqrt{15}+3$

34) $A=\sqrt{5}-\sqrt{3}$ $B=\sqrt{5}+\sqrt{3}$ $\frac{A}{B}+\frac{B}{A}+2=?$

$\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}+\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}+2=\frac{5-2\sqrt{15}+3+5+2\sqrt{15}+3+2}{2}=8+2=10$

$(\sqrt{5}+\sqrt{3})$